

Schema documentation for gpcML.xsd

may 12, 2026

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Namespace: ""

Schema(s)

Main schema gpcML.xsd

Namespace	No namespace
Properties	attribute form default: unqualified
	element form default: unqualified

Element(s)

Element gpcML

Namespace	No namespace
Annotations	Root element of the XSD
Diagram	
Type	gpcMLType
Properties	content: complex
Model	SummaryValues{0,1} , ContactList , Metadata , InstrumentInfoList{0,1} , RunTimeData , CalibrationData
Children	CalibrationData, ContactList, InstrumentInfoList, Metadata, RunTimeData, SummaryValues
Instance	<pre><gpcML id="" Version=""> <SummaryValues>{0,1}</SummaryValues> <ContactList>{1,1}</ContactList></pre>

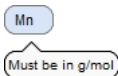
	<pre><Metadata CalibrationName="" date="" InstrumentName="" lastServiceDate="" MethodName="" Operator="" RunTime="" S Metadata> <InstrumentInfoList>{0,1}</InstrumentInfoList> <RunTimeData>{1,1}</RunTimeData> <CalibrationData>{1,1}</CalibrationData> </gpcML></pre>			
Attributes	QName	Type	Use	
	Version	xs:string	required	
		gpcML version used.		
	id	xs:int	optional	
		id associated with the gpcML document		
Source	<pre><xs:element name="gpcML" type="gpcMLType"> <xs:annotation> <xs:documentation>Root element of the XSD</xs:documentation> </xs:annotation> </xs:element></pre>			

Element gpcMLType / SummaryValues

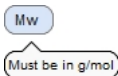
Namespace	No namespace
Diagram	SummaryValuesType (extension base) - Mn (Must be in g/mol) - Mw (Must be in g/mol) - PDI - Mz (Must be in g/mol) - Mv (Must be in g/mol) - n (Type: xs.float, Must be in ml/g) - Vp (Must be in ml) - Mp (Must be in g/mol) - A (Must be in ml*v) SummaryValues (Type: Extension of 'SummaryValuesType') Contains summary values for the polymer, it is best practise to ignore these values if recieving from
Type	extension of SummaryValuesType
Type hierarchy	SummaryValuesType
Properties	content: complex
	minOccurs: 0
Model	Mn{0,1} , Mw{0,1} , PDI{0,1} , Mz{0,1} , Mv{0,1} , n{0,1} , Vp{0,1} , Mp{0,1} , A{0,1}
Children	A, Mn, Mp, Mv, Mw, Mz, PDI, Vp, n
Instance	<pre> <SummaryValues> <Mn>{0,1}</Mn> <Mw>{0,1}</Mw> <PDI>{0,1}</PDI> <Mz>{0,1}</Mz> </pre>

	<pre> <Mv>{0,1}</Mv> <n>{0,1}</n> <Vp>{0,1}</Vp> <Mp>{0,1}</Mp> <A>{0,1} </SummaryValues> </pre>
Source	<pre> <xs:element name="SummaryValues" minOccurs="0"> <xs:complexType> <xs:complexContent> <xs:extension base="SummaryValuesType"/> </xs:complexContent> </xs:complexType> </xs:element> </pre>

Element SummaryValuesType / Mn

Namespace	No namespace
Annotations	Must be in g/mol
Diagram	
Properties	minOccurs: 0
Source	<pre> <xs:element name="Mn" minOccurs="0"> <xs:annotation> <xs:documentation>Must be in g/mol</xs:documentation> </xs:annotation> </xs:element> </pre>

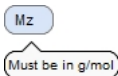
Element SummaryValuesType / Mw

Namespace	No namespace
Annotations	Must be in g/mol
Diagram	
Properties	minOccurs: 0
Source	<pre> <xs:element name="Mw" minOccurs="0"> <xs:annotation> <xs:documentation>Must be in g/mol</xs:documentation> </xs:annotation> </xs:element> </pre>

Element SummaryValuesType / PDI

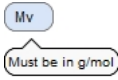
Namespace	No namespace
Diagram	
Properties	minOccurs: 0
Source	<pre> <xs:element name="PDI" minOccurs="0"/> </pre>

Element SummaryValuesType / Mz

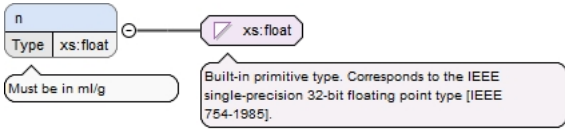
Namespace	No namespace
Annotations	Must be in g/mol
Diagram	
Properties	minOccurs: 0
Source	<pre> <xs:element name="Mz" minOccurs="0"> <xs:annotation> <xs:documentation>Must be in g/mol</xs:documentation> </xs:annotation> </xs:element> </pre>

```
</xs:annotation>
</xs:element>
```

Element SummaryValuesType / Mv

Namespace	No namespace
Annotations	Must be in g/mol
Diagram	
Properties	minOccurs: 0
Source	<pre><xs:element name="Mv" minOccurs="0"> <xs:annotation> <xs:documentation>Must be in g/mol</xs:documentation> </xs:annotation> </xs:element></pre>

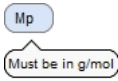
Element SummaryValuesType / n

Namespace	No namespace
Annotations	Must be in ml/g
Diagram	
Type	xs:float
Properties	content: simple minOccurs: 0
Source	<pre><xs:element name="n" type="xs:float" minOccurs="0"> <xs:annotation> <xs:documentation>Must be in ml/g</xs:documentation> </xs:annotation> </xs:element></pre>

Element SummaryValuesType / Vp

Namespace	No namespace
Annotations	Must be in ml
Diagram	
Properties	minOccurs: 0
Source	<pre><xs:element name="Vp" minOccurs="0"> <xs:annotation> <xs:documentation>Must be in ml</xs:documentation> </xs:annotation> </xs:element></pre>

Element SummaryValuesType / Mp

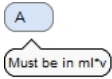
Namespace	No namespace
Annotations	Must be in g/mol
Diagram	
Properties	minOccurs: 0
Source	<pre><xs:element name="Mp" minOccurs="0"> <xs:annotation></pre>

```

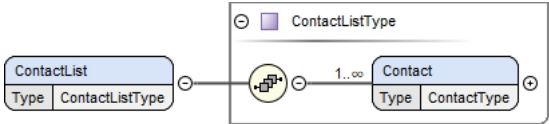
<xs:documentation>Must be in g/mol</xs:documentation>
</xs:annotation>
</xs:element>

```

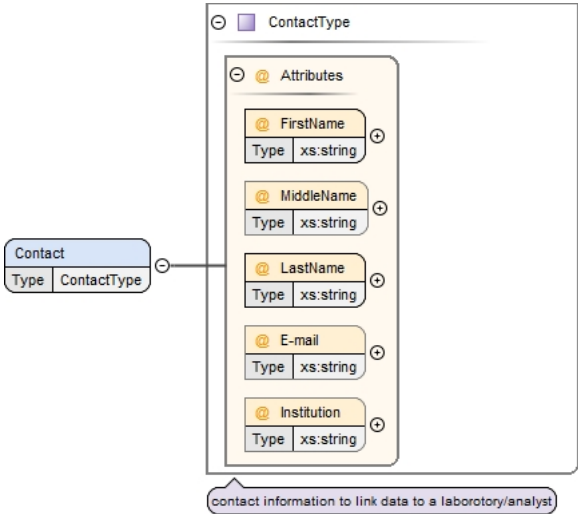
Element SummaryValuesType / A

Namespace	No namespace
Annotations	Must be in ml*v
Diagram	
Properties	minOccurs: 0
Source	<pre> <xs:element name="A" minOccurs="0"> <xs:annotation> <xs:documentation>Must be in ml*v</xs:documentation> </xs:annotation> </xs:element> </pre>

Element gpcMLType / ContactList

Namespace	No namespace
Diagram	
Type	ContactListType
Properties	content: complex
Model	Contact+
Children	Contact
Instance	<pre> <ContactList> <Contact E-mail="" FirstName="" Institution="" LastName="" MiddleName="">{1,unbounded}</Contact> </ContactList> </pre>
Source	<pre> <xs:element name="ContactList" type="ContactListType"/> </pre>

Element ContactListType / Contact

Namespace	No namespace
Diagram	
Type	ContactType
Properties	content: complex maxOccurs: unbounded

Attributes	QName	Type	Use	
	E-mail	xs:string	optional	
	FirstName	xs:string	required	
	Institution	xs:string	optional	
	LastName	xs:string	required	
	MiddleName	xs:string	optional	
Source	<xs:element name="Contact" type="ContactType" maxOccurs="unbounded"/>			

Element gpcMLType / Metadata

Namespace	No namespace			
Diagram				
Type	MetadataType			
Properties	content: complex			
Attributes	QName	Type	Use	
	CalibrationName	xs:string	optional	
	InstrumentName	xs:string	required	
	MethodName	xs:string	required	
	Operator	xs:string	optional	
	If multiple contacts, state name (first-last) of who ran the sample			
	RunTime	xs:time	optional	
	Length of the run sample in Minutes			
	SampleName	xs:string	required	
	date	xs:date	required	
	lastServiceDate	xs:date	optional	
Source	<xs:element name="Metadata" type="MetadataType"/>			

Element gpcMLType / InstrumentInfoList

Namespace	No namespace
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Diagram	
Type	InstrumentListType
Properties	content: complex minOccurs: 0
Model	RIDetectorInfo UVvisInfo{0,1} LightScatteringInfo{0,1} ViscometerInfo{0,1} ColumnInfo
Children	ColumnInfo, LightScatteringInfo, RIDetectorInfo, UVvisInfo, ViscometerInfo
Instance	<pre><InstrumentInfoList> <RIDetectorInfo>{1,1}</RIDetectorInfo> <UVvisInfo>{0,1}</UVvisInfo> <LightScatteringInfo>{0,1}</LightScatteringInfo> <ViscometerInfo>{0,1}</ViscometerInfo> <ColumnInfo>{1,1}</ColumnInfo> </InstrumentInfoList></pre>
Source	<code><xs:element name="InstrumentInfoList" type="InstrumentListType" minOccurs="0"/></code>

Element InstrumentListType / RIDetectorInfo

Namespace	No namespace
Diagram	
Type	RIDetectorInfoType
Properties	content: complex
Model	IDinfo
Children	IDinfo
Instance	<pre><RIDetectorInfo> <IDinfo Manufacturer="" Name="" SerialNumber="">{1,1}</IDinfo> </RIDetectorInfo></pre>
Source	<code><xs:element name="RIDetectorInfo" type="RIDetectorInfoType"/></code>

Element RIDetectorInfoType / IDinfo

Namespace	No namespace
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Diagram	The diagram illustrates the structure of the <code>InstrumentInfoType</code> class. It is composed of an <code>IDinfo</code> class (which is a specialization or instance of <code>InstrumentInfoType</code>) and an <code>Attributes</code> class. The <code>Attributes</code> class contains three attributes: <code>Name</code>, <code>SerialNumber</code>, and <code>Manufacturer</code>, all of which are of type <code>xs:string</code>. A note for <code>SerialNumber</code> states: "for saving what column were used, can be found on providers website, must be an unambiguous string which can be...". A note for <code>Manufacturer</code> states: "for saving what columns were used". A general note for the <code>InstrumentInfoType</code> class states: "Specific information relating to the equipment to allow easy lookup of the parts".																								
Type	InstrumentInfoType																								
Properties	content: complex																								
Attributes	<table><thead><tr><th>QName</th><th>Type</th><th>Use</th><th></th></tr></thead><tbody><tr><td>Manufacturer</td><td>xs:string</td><td>required</td><td></td></tr><tr><td></td><td colspan="3">for saving what columns were used</td></tr><tr><td>Name</td><td>xs:string</td><td>optional</td><td></td></tr><tr><td>SerialNumber</td><td>xs:string</td><td>optional</td><td></td></tr><tr><td></td><td colspan="3">for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.</td></tr></tbody></table>	QName	Type	Use		Manufacturer	xs:string	required			for saving what columns were used			Name	xs:string	optional		SerialNumber	xs:string	optional			for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.		
QName	Type	Use																							
Manufacturer	xs:string	required																							
	for saving what columns were used																								
Name	xs:string	optional																							
SerialNumber	xs:string	optional																							
	for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.																								
Source	<pre><xs:element name="IDinfo" type="InstrumentInfoType"/></pre>																								

Element InstrumentListType / UVvisInfo

Namespace	No namespace		
Diagram	<pre>classDiagram class UVvisInfoType { UVvisInfo IDinfo } class UVvisInfo { Type UVvisInfoType } class IDinfo { Type InstrumentInfoType } UVvisInfoType -- UVvisInfo UVvisInfoType -- IDinfo</pre>		
Type	UVvisInfoType		
Properties	content:	complex	
	minOccurs:	0	
Model	IDinfo		
Children	IDinfo		
Instance	<pre><UVvisInfo> <IDinfo Manufacturer="" Name="" SerialNumber="">{1,1}</IDinfo> </UVvisInfo></pre>		
Source	<pre><xs:element name="UVvisInfo" type="UVvisInfoType" minOccurs="0"/></pre>		

Element UVvisInfoType / IDinfo

Namespace	No namespace
-----------	--------------

Diagram																									
Type	InstrumentInfoType																								
Properties	content: complex																								
Attributes	<table><thead><tr><th>QName</th><th>Type</th><th>Use</th><th></th></tr></thead><tbody><tr><td>Manufacturer</td><td>xs:string</td><td>required</td><td></td></tr><tr><td></td><td colspan="3">for saving what columns were used</td></tr><tr><td>Name</td><td>xs:string</td><td>optional</td><td></td></tr><tr><td>SerialNumber</td><td>xs:string</td><td>optional</td><td></td></tr><tr><td></td><td colspan="3">for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.</td></tr></tbody></table>	QName	Type	Use		Manufacturer	xs:string	required			for saving what columns were used			Name	xs:string	optional		SerialNumber	xs:string	optional			for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.		
QName	Type	Use																							
Manufacturer	xs:string	required																							
	for saving what columns were used																								
Name	xs:string	optional																							
SerialNumber	xs:string	optional																							
	for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.																								
Source	<xs:element name="IDinfo" type="InstrumentInfoType"/>																								

Element InstrumentListType / LightScatteringInfo

Namespace	No namespace		
Diagram	<pre>classDiagram class LightScatteringInfoType { Type LightScatteringInfoType } class IDinfo { Type InstrumentInfoType } LightScatteringInfoType "1" -- "1" IDinfo</pre>		
Type	LightScatteringInfoType		
Properties	content:	complex	
	minOccurs:	0	
Model	IDinfo		
Children	IDinfo		
Instance	<pre><LightScatteringInfo> <IDinfo Manufacturer="" Name="" SerialNumber="">{1,1}</IDinfo> </LightScatteringInfo></pre>		
Source	<pre><xs:element name="LightScatteringInfo" type="LightScatteringInfoType" minOccurs="0"/></pre>		

Element LightScatteringInfoType / IDinfo

Namespace	No namespace
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Diagram	<pre>classDiagram class IDinfo { Type InstrumentInfoType } class InstrumentInfoType { Attributes class Attributes { Name Type xs:string SerialNumber Type xs:string Manufacturer Type xs:string } } IDinfo --> InstrumentInfoType</pre>																								
Type	InstrumentInfoType																								
Properties	content: complex																								
Attributes	<table><thead><tr><th>QName</th><th>Type</th><th>Use</th><th></th></tr></thead><tbody><tr><td>Manufacturer</td><td>xs:string</td><td>required</td><td></td></tr><tr><td></td><td colspan="3">for saving what columns were used</td></tr><tr><td>Name</td><td>xs:string</td><td>optional</td><td></td></tr><tr><td>SerialNumber</td><td>xs:string</td><td>optional</td><td></td></tr><tr><td></td><td colspan="3">for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.</td></tr></tbody></table>	QName	Type	Use		Manufacturer	xs:string	required			for saving what columns were used			Name	xs:string	optional		SerialNumber	xs:string	optional			for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.		
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Name	xs:string	optional																							
SerialNumber	xs:string	optional																							
	for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.																								
Source	<pre><xs:element name="IDinfo" type="InstrumentInfoType"/></pre>																								

Element InstrumentListType / ViscometerInfo

Namespace	No namespace		
Diagram	<pre>classDiagram class ViscometerInfoType { Type } class IDInfo { Type InstrumentInfoType } ViscometerInfoType "1" -- "*" IDInfo</pre>		
Type	ViscometerInfoType		
Properties	content:	complex	
	minOccurs:	0	
Model	IDInfo		
Children	IDInfo		
Instance	<pre><ViscometerInfo> <IDInfo Manufacturer="" Name="" SerialNumber="">{1,1}</IDInfo> </ViscometerInfo></pre>		
Source	<pre><xs:element name="ViscometerInfo" type="ViscometerInfoType" minOccurs="0"/></pre>		

Element ViscometerInfoType / IDinfo

Namespace	No namespace
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Diagram	The diagram shows a class-like structure for InstrumentInfoType. It has three attributes: Name (Type: xs:string), SerialNumber (Type: xs:string), and Manufacturer (Type: xs:string). A note for SerialNumber states: "for saving what column were used, can be found on providers website, must be an unambiguous string which can be...". A note for Manufacturer states: "for saving what columns were used". A general note at the bottom states: "Specific information relating to the equipment to allow easy lookup of the parts".																								
Type	InstrumentInfoType																								
Properties	content: complex																								
Attributes	<table><thead><tr><th>QName</th><th>Type</th><th>Use</th><th></th></tr></thead><tbody><tr><td>Manufacturer</td><td>xs:string</td><td>required</td><td></td></tr><tr><td></td><td colspan="3">for saving what columns were used</td></tr><tr><td>Name</td><td>xs:string</td><td>optional</td><td></td></tr><tr><td>SerialNumber</td><td>xs:string</td><td>optional</td><td></td></tr><tr><td></td><td colspan="3">for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.</td></tr></tbody></table>	QName	Type	Use		Manufacturer	xs:string	required			for saving what columns were used			Name	xs:string	optional		SerialNumber	xs:string	optional			for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.		
QName	Type	Use																							
Manufacturer	xs:string	required																							
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Name	xs:string	optional																							
SerialNumber	xs:string	optional																							
	for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.																								
Source	<pre><xs:element name="IDinfo" type="InstrumentInfoType"/></pre>																								

Element InstrumentListType / ColumnInfo

Namespace	No namespace			
Annotations	order of storage is what determines which columninfo is associated with which column in runtime data (i.e first instance of column info is related to the first instance of runtime data column)			
Diagram				
Type	ColumnInfoListType			
Properties	content: complex			
Model	IDinfo			
Children	IDinfo			
Instance	<pre><ColumnInfo> <IDinfo Manufacturer="" Name="" SerialNumber="">{1,1}</IDinfo> </ColumnInfo></pre>			
Source	<pre><xs:element name="ColumnInfo" type="ColumnInfoListType"> <xs:annotation> <xs:documentation>order of storage is what determines which columninfo is associated with which column in runtime data (i.e first instance of column info is related to the first instance of runtime data column</xs:documentation> </xs:annotation> </xs:element></pre>			

Element ColumnInfoListType / IDinfo

Namespace	No namespace
-----------	--------------

Diagram	The diagram illustrates the structure of the <code>InstrumentInfoType</code>. It is a complex type with three attributes: <code>Name</code> (type <code>xs:string</code>), <code>SerialNumber</code> (type <code>xs:string</code>), and <code>Manufacturer</code> (type <code>xs:string</code>). A note for <code>SerialNumber</code> states: "for saving what column were used, can be found on providers website, must be an unambiguous string which can be...". A note for <code>Manufacturer</code> states: "for saving what columns were used". A general note for the type states: "Specific information relating to the equipment to allow easy lookup of the parts".																								
Type	InstrumentInfoType																								
Properties	content: complex																								
Attributes	<table><thead><tr><th>QName</th><th>Type</th><th>Use</th><th></th></tr></thead><tbody><tr><td>Manufacturer</td><td>xs:string</td><td>required</td><td></td></tr><tr><td></td><td colspan="3">for saving what columns were used</td></tr><tr><td>Name</td><td>xs:string</td><td>optional</td><td></td></tr><tr><td>SerialNumber</td><td>xs:string</td><td>optional</td><td></td></tr><tr><td></td><td colspan="3">for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.</td></tr></tbody></table>	QName	Type	Use		Manufacturer	xs:string	required			for saving what columns were used			Name	xs:string	optional		SerialNumber	xs:string	optional			for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.		
QName	Type	Use																							
Manufacturer	xs:string	required																							
	for saving what columns were used																								
Name	xs:string	optional																							
SerialNumber	xs:string	optional																							
	for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.																								
Source	<pre><xs:element name="IDinfo" type="InstrumentInfoType"/></pre>																								

Element gpcMLType / RunTimeData

Namespace	No namespace			
Diagram				
Type	RuntimeDataType			
Properties	content: complex			
Model	InstrumentConfiguration , RawData , IntergrationValues			
Children	InstrumentConfiguration, IntergrationValues, RawData			
Instance	<pre><RunTimeData> <InstrumentConfiguration>{1,1}</InstrumentConfiguration> <RawData Compression="">{1,1}</RawData> <IntergrationValues>{1,1}</IntergrationValues> </RunTimeData></pre>			
Source	<pre><xs:element name="RunTimeData" type="RuntimeDataType"/></pre>			

Element RuntimeDataType / InstrumentConfiguration

Namespace	No namespace
-----------	--------------

Diagram	
Type	InstrumentConfigurationType
Properties	content: complex
Model	ColumnList , OvenList , Pump , Injector{0,1} , Eluent
Children	ColumnList, Eluent, Injector, OvenList, Pump
Instance	<pre> <InstrumentConfiguration> <ColumnList>{1,1}</ColumnList> <OvenList>{1,1}</OvenList> <Pump FlowRate="">{1,1}</Pump> <Injector InjectionVolume="">{0,1}</Injector> <Eluent>{1,1}</Eluent> </InstrumentConfiguration> </pre>
Source	<code><xs:element name="InstrumentConfiguration" type="InstrumentConfigurationType"/></code>

Element InstrumentConfigurationType / ColumnList

Namespace	No namespace
Annotations	Columns stored in the order the sample runs through, column at the top of the list will be the column which separates the sample first
Diagram	
Type	ColumnListConfigurationType
Properties	content: complex
Model	GuardColumn{0,1} , Column+
Children	Column, GuardColumn
Instance	<pre> <ColumnList> <GuardColumn DateOfInstallation="" PoreSizeMax="" PoreSizeMin="" StationaryPhase="">{0,1}</GuardColumn> <Column DateOfInstallation="" PoreSizeMax="" PoreSizeMin="" StationaryPhase="">{1,unbounded}</Column> </ColumnList> </pre>
Source	<pre> <xs:element name="ColumnList" type="ColumnListConfigurationType"> <xs:annotation> <xs:documentation>Columns stored in the order the sample runs through, column at the top of the list will be the column which separates the sample first</xs:documentation> </xs:annotation> </xs:element> </pre>

Element ColumnListConfigurationType / GuardColumn

Namespace	No namespace
-----------	--------------

Diagram				
Type	ColumnConfigurationType			
Properties	content:	complex		
	minOccurs:	0		
Attributes	QName	Type	Use	
	DateOfInstallation	xs:date	optional	
	PoreSizeMax	xs:positiveInteger	required	
	PoreSizeMin	xs:positiveInteger	required	
	StationaryPhase	xs:string	optional	
Source	<pre><xs:element name="GuardColumn" type="ColumnConfigurationType" minOccurs="0"/></pre>			

Element ColumnListConfigurationType / Column

Namespace	No namespace			
Annotations	Information should be looked up on the relevant providers website			
Diagram				
Type	ColumnConfigurationType			
Properties	content:	complex		
	maxOccurs:	unbounded		
Attributes	QName	Type	Use	
	DateOfInstallation	xs:date	optional	
	PoreSizeMax	xs:positiveInteger	required	
	PoreSizeMin	xs:positiveInteger	required	
	StationaryPhase	xs:string	optional	
Source	<pre><xs:element name="Column" type="ColumnConfigurationType" maxOccurs="unbounded"> <xs:annotation> <xs:documentation>Information should be looked up on the relevant providers website</xs:documentation> </xs:annotation> </xs:element></pre>			

Element InstrumentConfigurationType / OvenList

Namespace	No namespace
Diagram	
Type	OvenListType
Properties	content: complex
Model	ColumnOven gpcOven
Children	ColumnOven, gpcOven
Instance	<pre><OvenList> <ColumnOven Temperature="">{1,1}</ColumnOven> <gpcOven Temperature="">{1,1}</gpcOven> </OvenList></pre>
Source	<code><xs:element name="OvenList" type="OvenListType"/></code>

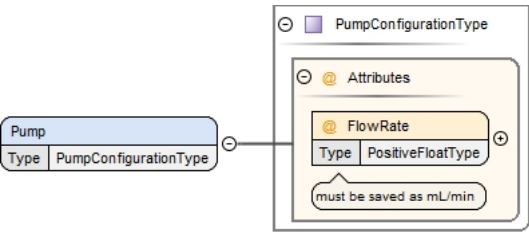
Element OvenListType / ColumnOven

Namespace	No namespace															
Diagram	The diagram illustrates the structure of the <code>ColumnOven</code> element. It is a container element of type <code>OvenConfigurationType</code>. Inside <code>OvenConfigurationType</code>, there is an <code>Attributes</code> container. Within <code>Attributes</code>, there is a <code>Temperature</code> element of type <code>PositiveFloatType</code>. A note indicates that the temperature must be saved in Celsius.															
Type	OvenConfigurationType															
Properties	content:	complex														
Attributes	<table><thead><tr><th>QName</th><th>Type</th><th>Use</th><th></th></tr></thead><tbody><tr><td>Temperature</td><td>PositiveFloatType</td><td>required</td><td></td></tr><tr><td colspan="4">Must be saved in celcius</td></tr></tbody></table>	QName	Type	Use		Temperature	PositiveFloatType	required		Must be saved in celcius						
QName	Type	Use														
Temperature	PositiveFloatType	required														
Must be saved in celcius																
Source	<code><xs:element name="ColumnOven" type="OvenConfigurationType"/></code>															

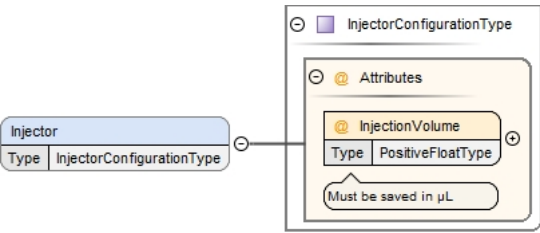
Element OvenListType / gpcOven

Namespace	No namespace			
Diagram	The diagram illustrates the structure of the <code>gpcOven</code> element. It is a container element of type <code>OvenConfigurationType</code>. Inside, it contains a single element of type <code>PositiveFloatType</code>, labeled <code>Temperature</code>. A note indicates that the value must be saved in Celsius.			
Type	OvenConfigurationType			
Properties	content:	complex		
Attributes	QName	Type	Use	
	Temperature	PositiveFloatType	required	
	Must be saved in celcius			
Source	<code><xs:element name="gpcOven" type="OvenConfigurationType"/></code>			

Element InstrumentConfigurationType / Pump

Namespace	No namespace			
Diagram				
Type	PumpConfigurationType			
Properties	content: complex			
Attributes	QName	Type	Use	
	FlowRate	PositiveFloatType	required	
	must be saved as mL/min			
Source	<pre><xs:element name="Pump" type="PumpConfigurationType"/></pre>			

Element InstrumentConfigurationType / Injector

Namespace	No namespace			
Diagram				
Type	InjectorConfigurationType			
Properties	content:	complex		
	minOccurs:	0		
Attributes	QName	Type	Use	
	InjectionVolume	PositiveFloatType	required	
	Must be saved in µL			
Source	<pre><xs:element name="Injector" type="InjectorConfigurationType" minOccurs="0"/></pre>			

Element InstrumentConfigurationType / Eluent

Namespace	No namespace			
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Diagram	
Type	EluentType
Properties	content: complex
Model	SolventName , AdditiveName {0,1} , RecyclingMode , (Ratio AdditiveConcentration)
Children	AdditiveConcentration, AdditiveName, Ratio, RecyclingMode, SolventName
Instance	<pre> <Eluent> <SolventName>{1,1}</SolventName> <AdditiveName>{0,1}</AdditiveName> <RecyclingMode>{1,1}</RecyclingMode> <Ratio>{1,1}</Ratio> <AdditiveConcentration>{1,1}</AdditiveConcentration> </Eluent> </pre>
Source	<code><xs:element name="Eluent" type="EluentType"/></code>

Element EluentType / SolventName

Namespace	No namespace
Diagram	
Type	xs:string
Properties	content: simple
Source	<code><xs:element name="SolventName" type="xs:string"/></code>

Element EluentType / AdditiveName

Namespace	No namespace
Diagram	
Type	xs:string
Properties	content: simple minOccurs: 0
Source	<code><xs:element name="AdditiveName" type="xs:string" minOccurs="0"/></code>

Element EluentType / RecyclingMode

Namespace	No namespace
-----------	--------------

Annotations	simple boolean value
Diagram	
Type	xs:boolean
Properties	content: simple
Source	<pre><xs:element name="RecyclingMode" type="xs:boolean"> <xs:annotation> <xs:documentation>simple boolean value</xs:documentation> </xs:annotation> </xs:element></pre>

Element EluentType / Ratio

Namespace	No namespace
Annotations	Ratio between the solvent and additive, if necessary. does not check if both sides add to 100.
Diagram	
Type	RatioType
Properties	content: simple
Facets	pattern (100 [0-9]?[0-9]):(100 [0-9]?[0-9])
Source	<pre><xs:element name="Ratio" type="RatioType"> <xs:annotation> <xs:documentation>Ratio between the solvent and additive, if necessary. does not check if both sides add to 100.</xs:documentation> </xs:annotation> </xs:element></pre>

Element EluentType / AdditiveConcentration

Namespace	No namespace
Annotations	concentration of additive in mg/L
Diagram	
Type	xs:float
Properties	content: simple
Source	<pre><xs:element name="AdditiveConcentration" type="xs:float"> <xs:annotation> <xs:documentation>concentration of additive in mg/L</xs:documentation> </xs:annotation> </xs:element></pre>

Element RuntimeDataType / RawData

Namespace	No namespace
Annotations	Stored as a base64 encoded data

Diagram				
Type	RawDataType			
Properties	content: complex			
Attributes	QName	Type	Use	
	Compression	xs:boolean	required	
Source	<pre><xs:element name="RawData" type="RawDataType"> <xs:annotation> <xs:documentation>Stored as a base64 encoded data</xs:documentation> </xs:annotation> </xs:element></pre>			

Element RuntimeDataType / IntergrationValues

Namespace	No namespace			
Diagram				
Type	IntergrationValuesListType			
Properties	content: complex			
Model	Baseline , Peaks+ , Solvent+			
Children	Baseline, Peaks, Solvent			
Instance	<pre><IntergrationValues> <Baseline EndVolume="" StartVolume="">{1,1}</Baseline> <Peaks EndVolume="" StartVolume="">{1,unbounded}</Peaks> <Solvent EndVolume="" StartVolume="">{1,unbounded}</Solvent> </IntergrationValues></pre>			
Source	<pre><xs:element name="IntergrationValues" type="IntergrationValuesListType"/></pre>			

Element IntergrationValuesListType / Baseline

Namespace	No namespace			
Diagram				
Type	IntergrationValuesType			

Properties	content: complex			
Attributes	QName	Type	Use	
	EndVolume	xs:float	required	
		Must be saved in mL		
	StartVolume	xs:float	required	
		Must be saved in mL		
Source	<xs:element name="Baseline" type="IntergrationValuesType"/>			

Element IntergrationValuesListType / Peaks

Namespace	No namespace			
Diagram				
Type	IntergrationValuesType			
Properties	content:	complex		
	maxOccurs:	unbounded		
Attributes	QName	Type	Use	
	EndVolume	xs:float	required	
		Must be saved in mL		
	StartVolume	xs:float	required	
		Must be saved in mL		
Source	<xs:element name="Peaks" type="IntergrationValuesType" maxOccurs="unbounded"/>			

Element IntergrationValuesListType / Solvent

Namespace	No namespace			
Diagram				
Type	IntergrationValuesType			
Properties	content:	complex		
	maxOccurs:	unbounded		
Attributes	QName	Type	Use	
	EndVolume	xs:float	required	
		Must be saved in mL		
	StartVolume	xs:float	required	

	QName	Type	Use	
		Must be saved in mL		
Source	<xs:element name="Solvent" type="IntergrationValuesType" maxOccurs="unbounded"/>			

Element gpcMLType / CalibrationData

Namespace	No namespace
Diagram	
Type	CalibrationDataType
Properties	content: complex
Model	TypeOfFit , CoeffList
Children	CoeffList, TypeOfFit
Instance	<pre><CalibrationData> <TypeOfFit>{1,1}</TypeOfFit> <CoeffList>{1,1}</CoeffList> </CalibrationData></pre>
Source	<xs:element name="CalibrationData" type="CalibrationDataType"/>

Element CalibrationDataType / TypeOfFit

Namespace	No namespace				
Annotations	Stores the degree of polynomial. used to affirm the correct number of coeffs are added.				
Diagram					
Type	DegreePolyType				
Properties	content: simple				
Facets	<table> <tr> <td>maxInclusive</td><td>5</td></tr> <tr> <td>minInclusive</td><td>3</td></tr> </table>	maxInclusive	5	minInclusive	3
maxInclusive	5				
minInclusive	3				
Source	<pre><xs:element name="TypeOfFit" type="DegreePolyType"> <xs:annotation> <xs:documentation>Stores the degree of polynomial. used to affirm the correct number of coeffs are added.</xs:documentation> </xs:annotation> </xs:element></pre>				

Element CalibrationDataType / CoeffList

Namespace	No namespace
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Diagram	
Type	CoeffListType
Properties	content: complex
Model	QuinticCoeff{0,1} , QuarticCoeff{0,1} , CubicCoeff , QuadraticCoeff , LinearCoeff , constant
Children	CubicCoeff, LinearCoeff, QuadraticCoeff, QuarticCoeff, QuinticCoeff, constant
Instance	<pre> <CoeffList> <QuinticCoeff>{0,1}</QuinticCoeff> <QuarticCoeff>{0,1}</QuarticCoeff> <CubicCoeff>{1,1}</CubicCoeff> <QuadraticCoeff>{1,1}</QuadraticCoeff> <LinearCoeff>{1,1}</LinearCoeff> <constant>{1,1}</constant> </CoeffList> </pre>
Source	<code><xs:element name="CoeffList" type="CoeffListType"/></code>

Element CoeffListType / QuinticCoeff

Namespace	No namespace
Diagram	
Type	CoeffType
Properties	content: simple
	minOccurs: 0
Source	<code><xs:element name="QuinticCoeff" type="CoeffType" minOccurs="0"/></code>

Element CoeffListType / QuarticCoeff

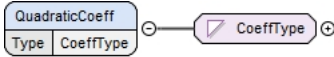
Namespace	No namespace
Diagram	
Type	CoeffType
Properties	content: simple
	minOccurs: 0
Source	<code><xs:element name="QuarticCoeff" type="CoeffType" minOccurs="0"/></code>

Element CoeffListType / CubicCoeff

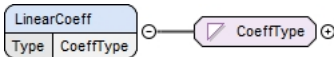
Namespace	No namespace
Diagram	

Type	CoeffType
Properties	content: simple
Source	<code><xs:element name="CubicCoeff" type="CoeffType"/></code>

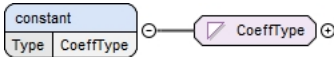
Element CoeffListType / QuadraticCoeff

Namespace	No namespace
Diagram	 <pre> graph LR subgraph QuadraticCoeff [QuadraticCoeff] direction TB Type[Type] CoeffType[CoeffType] end CoeffType </pre>
Type	CoeffType
Properties	content: simple
Source	<code><xs:element name="QuadraticCoeff" type="CoeffType"/></code>

Element CoeffListType / LinearCoeff

Namespace	No namespace
Diagram	 <pre> graph LR subgraph LinearCoeff [LinearCoeff] direction TB Type[Type] CoeffType[CoeffType] end CoeffType </pre>
Type	CoeffType
Properties	content: simple
Source	<code><xs:element name="LinearCoeff" type="CoeffType"/></code>

Element CoeffListType / constant

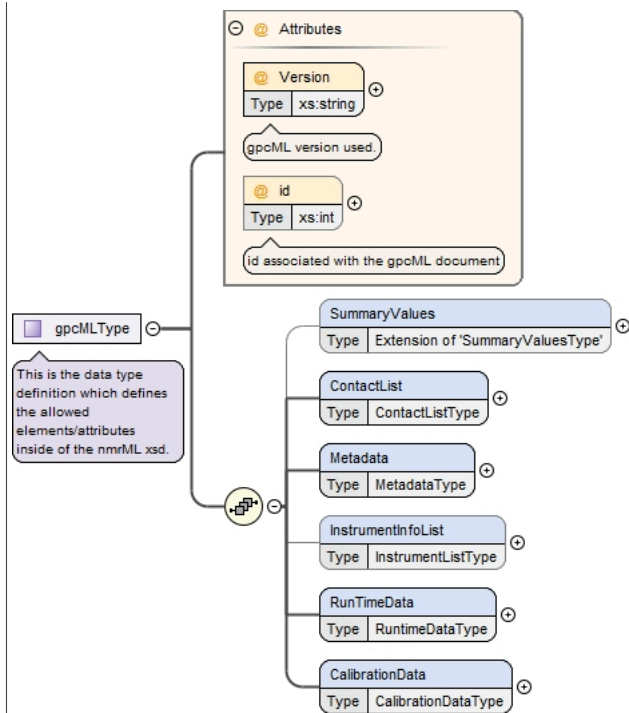
Namespace	No namespace
Diagram	 <pre> graph LR subgraph constant [constant] direction TB Type[Type] CoeffType[CoeffType] end CoeffType </pre>
Type	CoeffType
Properties	content: simple
Source	<code><xs:element name="constant" type="CoeffType"/></code>

Complex Type(s)

Complex Type gpcMLType

Namespace	No namespace
Annotations	This is the data type definition which defines the allowed elements/attributes inside of the nmrML xsd.

Diagram



Used by	Element gpcML			
Model	SummaryValues{0,1} , ContactList , Metadata , InstrumentInfoList{0,1} , RunTimeData , CalibrationData			
Children	CalibrationData, ContactList, InstrumentInfoList, Metadata, RunTimeData, SummaryValues			
Attributes	QName	Type	Use	
	Version	xs:string	required	
		gpcML version used.		
	id	xs:int	optional	
		id associated with the gpcML document		
Source	<pre><xs:complexType name="gpcMLType"> <xs:annotation> <xs:documentation>This is the data type definition which defines the allowed elements/attributes inside of the nmrML xsd.</xs:documentation> </xs:annotation> <xs:sequence> <xs:element name="SummaryValues" minOccurs="0"> <xs:complexType> <xs:complexContent> <xs:extension base="SummaryValuesType"/> </xs:complexContent> </xs:complexType> </xs:element> <xs:element name="ContactList" type="ContactListType"/> <xs:element name="Metadata" type="MetadataType"/> <xs:element name="InstrumentInfoList" type="InstrumentListType" minOccurs="0"/> <xs:element name="RunTimeData" type="RuntimeDataType"/> <xs:element name="CalibrationData" type="CalibrationDataType"/> </xs:sequence> <xs:attribute name="Version" type="xs:string" use="required"> <xs:annotation> <xs:documentation>gpcML version used.</xs:documentation> </xs:annotation> </xs:attribute> <xs:attribute name="id" type="xs:int"> <xs:annotation> <xs:documentation>id associated with the gpcML document</xs:documentation> </xs:annotation> </xs:attribute> </xs:complexType></pre>			

Complex Type SummaryValuesType

Namespace	No namespace
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Annotations	Contains summary values for the polymer, it is best practise to ignore these values if recieving from
Diagram	<pre> graph TD Root((SummaryValuesType)) --- Mn[Mn] Root --- Mw[Mw] Root --- PDI[PDI] Root --- Mz[Mz] Root --- Mv[Mv] Root --- n[n] Root --- Vp[Vp] Root --- Mp[Mp] Root --- A[A] Mn --- MnAnn[Must be in g/mol] Mw --- MwAnn[Must be in g/mol] Mz --- MzAnn[Must be in g/mol] Mv --- MvAnn[Must be in g/mol] n --- nAnn[Must be in ml/g] Vp --- VpAnn[Must be in ml] Mp --- MpAnn[Must be in g/mol] A --- AAnn[Must be in ml^v] n --- nType[Type: xs:float] </pre>
Used by	Element gpcMLType/SummaryValues
Model	Mn{0,1} , Mw{0,1} , PDI{0,1} , Mz{0,1} , Mv{0,1} , n{0,1} , Vp{0,1} , Mp{0,1} , A{0,1}
Children	A, Mn, Mp, Mv, Mw, Mz, PDI, Vp, n
Source	<pre> <xs:complexType name="SummaryValuesType"> <xs:annotation> <xs:documentation>Contains summary values for the polymer, it is best practise to ignore these values if recieving from</xs:documentation> </xs:annotation> <xs:sequence> <xs:element name="Mn" minOccurs="0"> <xs:annotation> <xs:documentation>Must be in g/mol</xs:documentation> </xs:annotation> </xs:element> <xs:element name="Mw" minOccurs="0"> <xs:annotation> <xs:documentation>Must be in g/mol</xs:documentation> </xs:annotation> </xs:element> <xs:element name="PDI" minOccurs="0"/> <xs:element name="Mz" minOccurs="0"> <xs:annotation> <xs:documentation>Must be in g/mol</xs:documentation> </xs:annotation> </xs:element> <xs:element name="Mv" minOccurs="0"> <xs:annotation> <xs:documentation>Must be in g/mol</xs:documentation> </xs:annotation> </xs:element> <xs:element name="n" type="xs:float" minOccurs="0"> <xs:annotation> <xs:documentation>Must be in ml/g</xs:documentation> </xs:annotation> </xs:element> <xs:element name="Vp" minOccurs="0"> <xs:annotation> <xs:documentation>Must be in ml</xs:documentation> </xs:annotation> </xs:element> <xs:element name="Mp" minOccurs="0"> <xs:annotation> </pre>

```

<xs:documentation>Must be in g/mol</xs:documentation>
</xs:annotation>
</xs:element>
<xs:element name="A" minOccurs="0">
  <xs:annotation>
    <xs:documentation>Must be in ml*v</xs:documentation>
  </xs:annotation>
</xs:element>
</xs:sequence>
</xs:complexType>

```

Complex Type ContactListType

Namespace	No namespace
Diagram	
Used by	Element gpcMLType/ContactList
Model	Contact+
Children	Contact
Source	<pre> <xs:complexType name="ContactListType"> <xs:sequence> <xs:element name="Contact" type="ContactType" maxOccurs="unbounded"/> </xs:sequence> </xs:complexType> </pre>

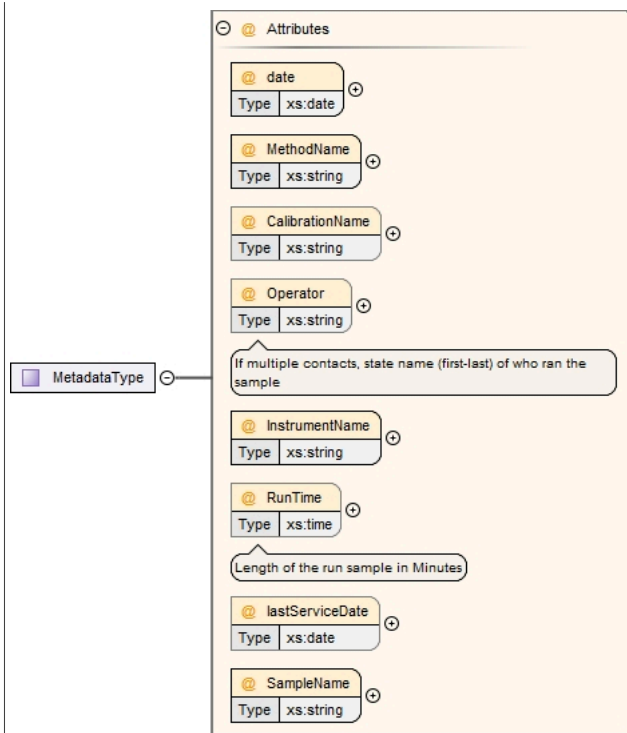
Complex Type ContactType

Namespace	No namespace																											
Annotations	contact information to link data to a laboratory/analyst																											
Diagram																												
Used by	Element ContactListType/Contact																											
Attributes	<table><thead><tr><th>QName</th><th>Type</th><th>Use</th><th></th></tr></thead><tbody><tr><td>E-mail</td><td>xs:string</td><td>optional</td><td></td></tr><tr><td>FirstName</td><td>xs:string</td><td>required</td><td></td></tr><tr><td>Institution</td><td>xs:string</td><td>optional</td><td></td></tr><tr><td>LastName</td><td>xs:string</td><td>required</td><td></td></tr><tr><td>MiddleName</td><td>xs:string</td><td>optional</td><td></td></tr></tbody></table>	QName	Type	Use		E-mail	xs:string	optional		FirstName	xs:string	required		Institution	xs:string	optional		LastName	xs:string	required		MiddleName	xs:string	optional				
QName	Type	Use																										
E-mail	xs:string	optional																										
FirstName	xs:string	required																										
Institution	xs:string	optional																										
LastName	xs:string	required																										
MiddleName	xs:string	optional																										
Source	<pre><xs:complexType name="ContactType"> <xs:annotation> <xs:documentation>contact information to link data to a laboratory/analyst</xs:documentation> </xs:annotation> <xs:attribute name="FirstName" type="xs:string" use="required"/> <xs:attribute name="MiddleName" type="xs:string" use="optional"/> <xs:attribute name="LastName" type="xs:string" use="required"/> <xs:attribute name="E-mail" type="xs:string"/> <xs:attribute name="Institution" type="xs:string" use="optional"/> </xs:complexType></pre>																											

Complex Type MetadataType

Namespace	No namespace
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Diagram

Used by
Element gpcMLType/Metadata

Attributes	QName	Type	Use	
	CalibrationName	xs:string	optional	
	InstrumentName	xs:string	required	
	MethodName	xs:string	required	
	Operator	xs:string	optional	
		If multiple contacts, state name (first-last) of who ran the sample		
	RunTime	xs:time	optional	
		Length of the run sample in Minutes		
	SampleName	xs:string	required	
	date	xs:date	required	
	lastServiceDate	xs:date	optional	

Source

```

<xs:complexType name="MetadataType">
  <xs:attribute name="date" type="xs:date" use="required"/>
  <xs:attribute name="MethodName" type="xs:string" use="required"/>
  <xs:attribute name="CalibrationName" type="xs:string" use="optional"/>
  <xs:attribute name="Operator" type="xs:string">
    <xs:annotation>
      <xs:documentation>If multiple contacts, state name (first-last) of who ran the sample</xs:documentation>
    </xs:annotation>
  </xs:attribute>
  <xs:attribute name="InstrumentName" type="xs:string" use="required"/>
  <xs:attribute name="RunTime" type="xs:time" use="optional">
    <xs:annotation>
      <xs:documentation>Length of the run sample in Minutes</xs:documentation>
    </xs:annotation>
  </xs:attribute>
  <xs:attribute name="lastServiceDate" type="xs:date" use="optional"/>
  <xs:attribute name="SampleName" type="xs:string" use="required"/>
</xs:complexType>

```

Complex Type InstrumentListType

Namespace	No namespace
Annotations	A complexType to contain all necessary instruments used to produce the experimental data

Diagram	
Used by	Element gpcMLType/InstrumentInfoList
Model	RIDetectorInfo UVvisInfo{0,1} LightScatteringInfo{0,1} ViscometerInfo{0,1} ColumnInfo
Children	ColumnInfo, LightScatteringInfo, RIDetectorInfo, UVvisInfo, ViscometerInfo
Source	<pre> <xs:complexType name="InstrumentListType"> <xs:annotation> <xs:documentation>A complexType to contain all necessary instruments used to produce the experimental data</xs:documentation> </xs:annotation> <xs:choice maxOccurs="unbounded"> <xs:element name="RIDetectorInfo" type="RIDetectorInfoType"/> <xs:element name="UVvisInfo" type="UVvisInfoType" minOccurs="0"/> <xs:element name="LightScatteringInfo" type="LightScatteringInfoType" minOccurs="0"/> <xs:element name="ViscometerInfo" type="ViscometerInfoType" minOccurs="0"/> <xs:element name="ColumnInfo" type="ColumnInfoListType"> <xs:annotation> <xs:documentation>order of storage is what determines which columninfo is associated with which column in runtime data (i.e first instance of column info is related to the first instance of runtime data column</xs:documentation> </xs:annotation> </xs:element> </xs:choice> </xs:complexType> </pre>

Complex Type RIDetectorInfoType

Namespace	No namespace
Diagram	
Used by	Element InstrumentListType/RIDetectorInfo
Model	IDinfo
Children	IDinfo
Source	<pre> <xs:complexType name="RIDetectorInfoType"> <xs:sequence> <xs:element name="IDinfo" type="InstrumentInfoType"/> </xs:sequence> </xs:complexType> </pre>

Complex Type InstrumentInfoType

Namespace	No namespace
Annotations	Specific information relating to the equipment to allow easy lookup of the parts

Diagram	InstrumentInfoType Specific information relating to the equipment to allow easy lookup of the parts Attributes Name Typexs:string SerialNumber Typexs:string for saving what column were used, can be found on providers website, must be an unambiguous string which can be... Manufacturer Typexs:string for saving what columns were used																											
Used by	Elements	ColumnInfoListType/IDinfo, LightScatteringInfoType/IDinfo, RIDetectorInfoType/IDinfo, UVvisInfoType/IDinfo, ViscometerInfoType/IDinfo																										
Attributes	<table><thead><tr><th>QName</th><th>Type</th><th>Use</th><th></th></tr></thead><tbody><tr><td>Manufacturer</td><td>xs:string</td><td>required</td><td></td></tr><tr><td></td><td colspan="3">for saving what columns were used</td></tr><tr><td>Name</td><td>xs:string</td><td>optional</td><td></td></tr><tr><td>SerialNumber</td><td>xs:string</td><td>optional</td><td></td></tr><tr><td></td><td colspan="3">for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.</td></tr></tbody></table>	QName	Type	Use		Manufacturer	xs:string	required			for saving what columns were used			Name	xs:string	optional		SerialNumber	xs:string	optional			for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.					
QName	Type	Use																										
Manufacturer	xs:string	required																										
	for saving what columns were used																											
Name	xs:string	optional																										
SerialNumber	xs:string	optional																										
	for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.																											
Source	<pre><xs:complexType name="InstrumentInfoType"> <xs:annotation> <xs:documentation>Specific information relating to the equipment to allow easy lookup of the parts</xs:documentation> </xs:annotation> <xs:attribute name="Name" type="xs:string" use="optional"/> <xs:attribute name="SerialNumber" type="xs:string"> <xs:annotation> <xs:documentation>for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.</xs:documentation> </xs:annotation> </xs:attribute> <xs:attribute name="Manufacturer" type="xs:string" use="required"> <xs:annotation> <xs:documentation>for saving what columns were used</xs:documentation> </xs:annotation> </xs:attribute> </xs:complexType></pre>																											

Complex Type UVvisInfoType

Namespace	No namespace		
Diagram	The diagram shows a sequence container (a circle with a plus sign) containing two elements. The first element is a rectangle labeled 'UVvisInfoType' with a small square icon. The second element is a rounded rectangle labeled 'IDinfo' with a small circle icon. Below 'IDinfo' is a box labeled 'InstrumentInfoType'.		
Used by	Element	InstrumentListType/UVvisInfo	
Model	IDinfo		
Children	IDinfo		
Source	<pre><xs:complexType name="UVvisInfoType"> <xs:sequence> <xs:element name="IDinfo" type="InstrumentInfoType"/> </xs:sequence> </xs:complexType></pre>		

Complex Type LightScatteringInfoType

Namespace	No namespace			
Diagram				
Used by	Element	InstrumentListType/LightScatteringInfo		

Model	IDinfo
Children	IDinfo
Source	<pre><xs:complexType name="LightScatteringInfoType"> <xs:sequence> <xs:element name="IDinfo" type="InstrumentInfoType"/> </xs:sequence> </xs:complexType></pre>

Complex Type ViscometerInfoType

Namespace	No namespace
Diagram	
Used by	Element InstrumentListType/ViscometerInfo
Model	IDinfo
Children	IDinfo
Source	<pre><xs:complexType name="ViscometerInfoType"> <xs:sequence> <xs:element name="IDinfo" type="InstrumentInfoType"/> </xs:sequence> </xs:complexType></pre>

Complex Type ColumnInfoListType

Namespace	No namespace
Diagram	
Used by	Element InstrumentListType/ColumnInfo
Model	IDinfo
Children	IDinfo
Source	<pre><xs:complexType name="ColumnInfoListType"> <xs:sequence> <xs:element name="IDinfo" type="InstrumentInfoType"/> </xs:sequence> </xs:complexType></pre>

Complex Type RuntimeDataType

Namespace	No namespace
Annotations	Contains a list of: the elugram data in a binary heap, List of intergration volumes (if multiple intergrations exist), system peak and internal standard peak volumes
Diagram	
Used by	Element gpcMLType/RunTimeData
Model	InstrumentConfiguration , RawData , IntergrationValues
Children	InstrumentConfiguration, IntergrationValues, RawData
Source	<pre><xs:complexType name="RuntimeDataType"> <xs:annotation> <xs:documentation>Contains a list of: the elugram data in a binary heap, List of intergration volumes (if multiple intergrations exist), system peak and internal standard peak volumes</ xs:documentation> </xs:annotation></pre>

```

<xs:sequence>
  <xs:element name="InstrumentConfiguration" type="InstrumentConfigurationType"/>
  <xs:element name="RawData" type="RawDataType">
    <xs:annotation>
      <xs:documentation>Stored as a base64 encoded data</xs:documentation>
    </xs:annotation>
  </xs:element>
  <xs:element name="IntergrationValues" type="IntergrationValuesListType"/>
</xs:sequence>
</xs:complexType>

```

Complex Type InstrumentConfigurationType

Namespace	No namespace
Diagram	
Used by	Element RuntimeDataType/InstrumentConfiguration
Model	ColumnList , OvenList , Pump , Injector{0,1} , Eluent
Children	ColumnList, Eluent, Injector, OvenList, Pump
Source	<pre> <xs:complexType name="InstrumentConfigurationType"> <xs:sequence> <xs:element name="ColumnList" type="ColumnListConfigurationType"> <xs:annotation> <xs:documentation>Columns stored in the order the sample runs through, column at the top of the list will be the column which separates the sample first</xs:documentation> </xs:annotation> </xs:element> <xs:element name="OvenList" type="OvenListType"/> <xs:element name="Pump" type="PumpConfigurationType"/> <xs:element name="Injector" type="InjectorConfigurationType" minOccurs="0"/> <xs:element name="Eluent" type="EluentType"/> </xs:sequence> </xs:complexType> </pre>

Complex Type ColumnListConfigurationType

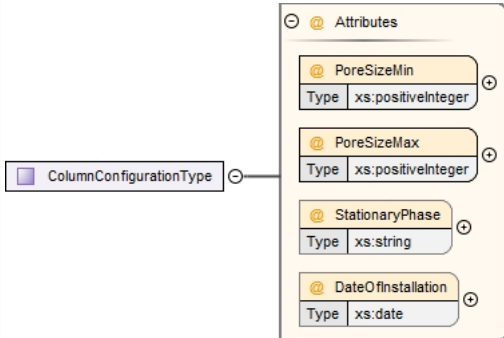
Namespace	No namespace
Diagram	
Used by	Element InstrumentConfigurationType/ColumnList
Model	GuardColumn{0,1} , Column+
Children	Column, GuardColumn
Source	<pre> <xs:complexType name="ColumnListConfigurationType"> <xs:sequence> <xs:element name="GuardColumn" type="ColumnConfigurationType" minOccurs="0"/> <xs:element name="Column" type="ColumnConfigurationType" maxOccurs="unbounded"> <xs:annotation> <xs:documentation>Information should be looked up on the relevant providers website</ xs:documentation> </xs:annotation> </xs:element> </xs:sequence> </xs:complexType> </pre>

```

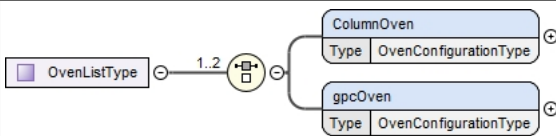
</xs:annotation>
</xs:element>
</xs:sequence>
</xs:complexType>

```

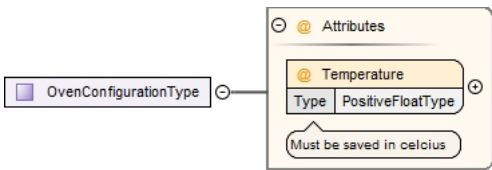
Complex Type ColumnConfigurationType

Namespace	No namespace																							
Diagram																								
Used by	Elements	ColumnListConfigurationType/Column, ColumnListConfigurationType/GuardColumn																						
Attributes	<table><thead><tr><th>QName</th><th>Type</th><th>Use</th><th></th></tr></thead><tbody><tr><td>DateOfInstallation</td><td>xs:date</td><td>optional</td><td></td></tr><tr><td>PoreSizeMax</td><td>xs:positiveInteger</td><td>required</td><td></td></tr><tr><td>PoreSizeMin</td><td>xs:positiveInteger</td><td>required</td><td></td></tr><tr><td>StationaryPhase</td><td>xs:string</td><td>optional</td><td></td></tr></tbody></table>	QName	Type	Use		DateOfInstallation	xs:date	optional		PoreSizeMax	xs:positiveInteger	required		PoreSizeMin	xs:positiveInteger	required		StationaryPhase	xs:string	optional				
QName	Type	Use																						
DateOfInstallation	xs:date	optional																						
PoreSizeMax	xs:positiveInteger	required																						
PoreSizeMin	xs:positiveInteger	required																						
StationaryPhase	xs:string	optional																						
Source	<pre><xs:complexType name="ColumnConfigurationType"> <xs:attribute name="PoreSizeMin" type="xs:positiveInteger" use="required"/> <xs:attribute name="PoreSizeMax" type="xs:positiveInteger" use="required"/> <xs:attribute name="StationaryPhase" type="xs:string" use="optional"/> <xs:attribute name="DateOfInstallation" type="xs:date" use="optional"/> </xs:complexType></pre>																							

Complex Type OvenListType

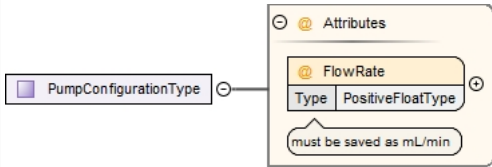
Namespace	No namespace
Diagram	
Used by	Element InstrumentConfigurationType/OvenList
Model	ColumnOven gpcOven
Children	ColumnOven, gpcOven
Source	<pre><xs:complexType name="OvenListType"> <xs:choice maxOccurs="2"> <xs:element name="ColumnOven" type="OvenConfigurationType"/> <xs:element name="gpcOven" type="OvenConfigurationType"/> </xs:choice> </xs:complexType></pre>

Complex Type OvenConfigurationType

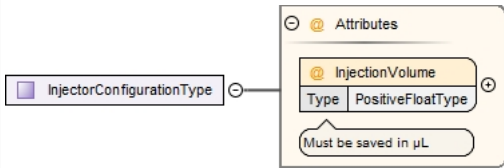
Namespace	No namespace			
Diagram				

Used by	ElementsOvenListType/ColumnOven, OvenListType/gpcOven			
Attributes	QName	Type	Use	
	Temperature	PositiveFloatType	required	
		Must be saved in celcius		
Source	<xs:complexType name="OvenConfigurationType"> <xs:attribute name="Temperature" type="PositiveFloatType" use="required"> <xs:annotation> <xs:documentation>Must be saved in celcius</xs:documentation> </xs:annotation> </xs:attribute> </xs:complexType>			

Complex Type PumpConfigurationType

Namespace	No namespace			
Diagram				
Used by	Element InstrumentConfigurationType/Pump			
Attributes	QName	Type	Use	
	FlowRate	PositiveFloatType	required	
		must be saved as mL/min		
Source	<pre><xs:complexType name="PumpConfigurationType"> <xs:attribute name="FlowRate" type="PositiveFloatType" use="required"> <xs:annotation> <xs:documentation>must be saved as mL/min</xs:documentation> </xs:annotation> </xs:attribute> </xs:complexType></pre>			

Complex Type InjectorConfigurationType

Namespace	No namespace			
Diagram				
Used by	Element InstrumentConfigurationType/Injector			
Attributes	QName	Type	Use	
	InjectionVolume	PositiveFloatType	required	
		Must be saved in µL		
Source	<pre><xs:complexType name="InjectorConfigurationType"> <xs:attribute name="InjectionVolume" type="PositiveFloatType" use="required"> <xs:annotation> <xs:documentation>Must be saved in µL</xs:documentation> </xs:annotation> </xs:attribute> </xs:complexType></pre>			

Complex Type EluentType

Namespace	No namespace			
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Diagram	
Used by	Element InstrumentConfigurationType/Eluent
Model	SolventName , AdditiveName{0,1} , RecyclingMode , (Ratio AdditiveConcentration)
Children	AdditiveConcentration, AdditiveName, Ratio, RecyclingMode, SolventName
Source	<pre> <xs:complexType name="EluentType"> <xs:sequence> <xs:element name="SolventName" type="xs:string"/> <xs:element name="AdditiveName" type="xs:string" minOccurs="0"/> <xs:element name="RecyclingMode" type="xs:boolean"> <xs:annotation> <xs:documentation>simple boolean value</xs:documentation> </xs:annotation> </xs:element> <xs:choice minOccurs="1" maxOccurs="2"> <xs:element name="Ratio" type="RatioType"> <xs:annotation> <xs:documentation>Ratio between the solvent and additive, if necessary. does not check if both sides add to 100.</xs:documentation> </xs:annotation> </xs:element> <xs:element name="AdditiveConcentration" type="xs:float"> <xs:annotation> <xs:documentation>concentration of additive in mg/L</xs:documentation> </xs:annotation> </xs:element> </xs:choice> </xs:sequence> </xs:complexType> </pre>

Complex Type RawDataType

Namespace	No namespace			
Diagram				
Type	extension of xs:base64Binary			
Used by	Element	RuntimeDataType/RawData		
Attributes	QName	Type	Use	
	Compression	xs:boolean	required	
Source	<pre> <xs:complexType name="RawDataType"> <xs:simpleContent> <xs:extension base="xs:base64Binary"> <xs:attribute name="Compression" type="xs:boolean" use="required"/> </xs:extension> </xs:simpleContent> </xs:complexType> </pre>			

```

</xs:simpleContent>
</xs:complexType>

```

Complex Type IntergrationValuesListType

Namespace	No namespace
Diagram	<pre> graph LR IVL[IntergrationValuesListType] --> B[Baseline] IVL --> P[Peaks] IVL --> S[Solvent] B --> B_Type[IntergrationValuesType] P --> P_Type[IntergrationValuesType] S --> S_Type[IntergrationValuesType] P --- P_Card[1..∞] S --- S_Card[1..∞] </pre>
Used by	Element RuntimeDataType/IntergrationValues
Model	Baseline , Peaks+ , Solvent+
Children	Baseline, Peaks, Solvent
Source	<pre> <xs:complexType name="IntergrationValuesListType"> <xs:sequence> <xs:element name="Baseline" type="IntergrationValuesType"/> <xs:element name="Peaks" type="IntergrationValuesType" maxOccurs="unbounded"/> <xs:element name="Solvent" type="IntergrationValuesType" maxOccurs="unbounded"/> </xs:sequence> </xs:complexType> </pre>

Complex Type IntergrationValuesType

Namespace	No namespace																							
Diagram	<pre>graph LR IVT[IntergrationValuesType] --> StartVolume IVT --> EndVolume StartVolume --> StartVolume_Type[xs:float] EndVolume --> EndVolume_Type[xs:float] StartVolume --- StartVolume_Doc[Must be saved in mL] EndVolume --- EndVolume_Doc[Must be saved in mL]</pre>																							
Used by	Elements	IntergrationValuesListType/Baseline, IntergrationValuesListType/Peaks, IntergrationValuesList-Type/Solvent																						
Attributes	<table><thead><tr><th>QName</th><th>Type</th><th>Use</th><th></th></tr></thead><tbody><tr><td>EndVolume</td><td>xs:float</td><td>required</td><td></td></tr><tr><td></td><td colspan="3">Must be saved in mL</td></tr><tr><td>StartVolume</td><td>xs:float</td><td>required</td><td></td></tr><tr><td></td><td colspan="3">Must be saved in mL</td></tr></tbody></table>	QName	Type	Use		EndVolume	xs:float	required			Must be saved in mL			StartVolume	xs:float	required			Must be saved in mL					
QName	Type	Use																						
EndVolume	xs:float	required																						
	Must be saved in mL																							
StartVolume	xs:float	required																						
	Must be saved in mL																							
Source	<pre><xs:complexType name="IntergrationValuesType"> <xs:attribute name="StartVolume" type="xs:float" use="required"> <xs:annotation> <xs:documentation>Must be saved in mL</xs:documentation> </xs:annotation> </xs:attribute> <xs:attribute name="EndVolume" type="xs:float" use="required"> <xs:annotation> <xs:documentation>Must be saved in mL</xs:documentation> </xs:annotation> </xs:attribute> </xs:complexType></pre>																							

Complex Type CaLibrationDataType

Namespace	No namespace
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Diagram	
Used by	Element gpcMLType/CalibrationData
Model	TypeOfFit , CoeffList
Children	CoeffList, TypeOfFit
Source	<pre> <xs:complexType name="CalibrationDataType"> <xs:sequence> <xs:element name="TypeOfFit" type="DegreePolyType"> <xs:annotation> <xs:documentation>Stores the degree of polynomial. used to affirm the correct number of coeffs are added.</xs:documentation> </xs:annotation> </xs:element> <xs:element name="CoeffList" type="CoeffListType"/> </xs:sequence> </xs:complexType> </pre>

Complex Type CoeffListType

Namespace	No namespace
Diagram	
Used by	Element CalibrationDataType/CoeffList
Model	QuinticCoeff{0,1} , QuarticCoeff{0,1} , CubicCoeff , QuadraticCoeff , LinearCoeff , constant
Children	CubicCoeff, LinearCoeff, QuadraticCoeff, QuarticCoeff, QuinticCoeff, constant
Source	<pre> <xs:complexType name="CoeffListType"> <xs:sequence> <xs:element name="QuinticCoeff" type="CoeffType" minOccurs="0"/> <xs:element name="QuarticCoeff" type="CoeffType" minOccurs="0"/> <xs:element name="CubicCoeff" type="CoeffType"/> <xs:element name="QuadraticCoeff" type="CoeffType"/> <xs:element name="LinearCoeff" type="CoeffType"/> <xs:element name="constant" type="CoeffType"/> </xs:sequence> </xs:complexType> </pre>

Complex Type Sample

Namespace	No namespace			
Diagram				
Attributes	QName	Type	Use	
	Name	xs:string	required	

Source	<pre><xs:complexType name="Sample"> <xs:attribute name="Name" type="xs:string" use="required"/> </xs:complexType></pre>
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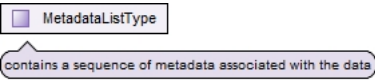
Complex Type SampleType

Namespace	No namespace
Diagram	
Source	<pre><xs:complexType name="SampleType"/></pre>

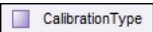
Complex Type EluGramType

Namespace	No namespace
Diagram	
Source	<pre><xs:complexType name="EluGramType"/></pre>

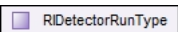
Complex Type MetadataListType

Namespace	No namespace
Annotations	contains a sequence of metadata associated with the data
Diagram	
Source	<pre><xs:complexType name="MetadataListType"> <xs:annotation> <xs:documentation>contains a sequence of metadata associated with the data</xs:documentation> </xs:annotation> </xs:complexType></pre>

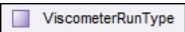
Complex Type CalibrationType

Namespace	No namespace
Diagram	
Source	<pre><xs:complexType name="CalibrationType"/></pre>

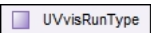
Complex Type RIDetectorRunType

Namespace	No namespace
Diagram	
Source	<pre><xs:complexType name="RIDetectorRunType"/></pre>

Complex Type ViscometerRunType

Namespace	No namespace
Diagram	
Source	<pre><xs:complexType name="ViscometerRunType"/></pre>

Complex Type UVvisRunType

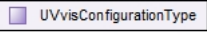
Namespace	No namespace
Diagram	
Source	<pre><xs:complexType name="UVvisRunType"/></pre>

Complex Type LightScatteringRunType

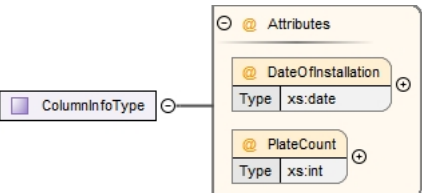
Namespace	No namespace
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Diagram	
Source	<code><xs:complexType name="LightScatteringRunType"/></code>

Complex Type UVvisConfigurationType

Namespace	No namespace
Diagram	
Source	<code><xs:complexType name="UVvisConfigurationType"/></code>

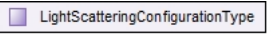
Complex Type ColumnInfoType

Namespace	No namespace												
Diagram													
Attributes	<table><thead><tr><th>QName</th><th>Type</th><th>Use</th><th></th></tr></thead><tbody><tr><td>DateOfInstallation</td><td>xs:date</td><td>optional</td><td></td></tr><tr><td>PlateCount</td><td>xs:int</td><td>optional</td><td></td></tr></tbody></table>	QName	Type	Use		DateOfInstallation	xs:date	optional		PlateCount	xs:int	optional	
QName	Type	Use											
DateOfInstallation	xs:date	optional											
PlateCount	xs:int	optional											
Source	<pre><xs:complexType name="ColumnInfoType"> <xs:attribute name="DateOfInstallation" type="xs:date" use="optional"/> <xs:attribute name="PlateCount" type="xs:int" use="optional"/> </xs:complexType></pre>												

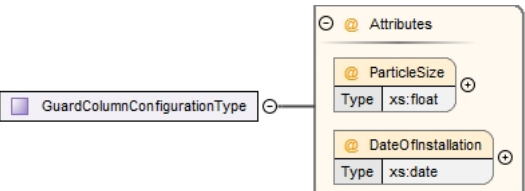
Complex Type RIDetectorConfigurationType

Namespace	No namespace
Diagram	
Source	<code><xs:complexType name="RIDetectorConfigurationType"/></code>

Complex Type LightScatteringConfigurationType

Namespace	No namespace
Diagram	
Source	<code><xs:complexType name="LightScatteringConfigurationType"/></code>

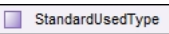
Complex Type GuardColumnConfigurationType

Namespace	No namespace			
Diagram				
Attributes	QName	Type	Use	
	DateOfInstallation	xs:date	optional	
	ParticleSize	xs:float	optional	
Source	<pre><xs:complexType name="GuardColumnConfigurationType"> <xs:attribute name="ParticleSize" type="xs:float" use="optional"/> <xs:attribute name="DateOfInstallation" type="xs:date" use="optional"/> </xs:complexType></pre>			

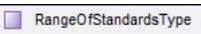
Complex Type PolyCoeffType

Namespace	No namespace
Diagram	
Source	<code><xs:complexType name="PolyCoeffType"/></code>

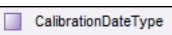
Complex Type StandardUsedType

Namespace	No namespace
Diagram	
Source	<code><xs:complexType name="StandardUsedType"/></code>

Complex Type RangeOfStandardsType

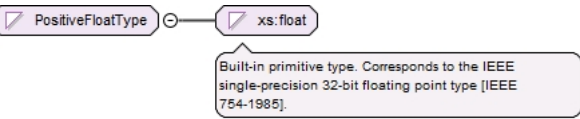
Namespace	No namespace
Diagram	
Source	<code><xs:complexType name="RangeOfStandardsType"/></code>

Complex Type CalibrationDateType

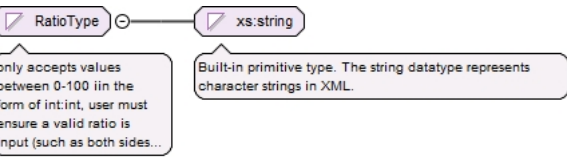
Namespace	No namespace
Diagram	
Source	<code><xs:complexType name="CalibrationDateType"/></code>

Simple Type(s)

Simple Type PositiveFloatType

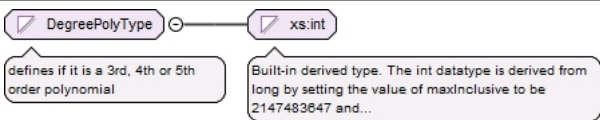
Namespace	No namespace
Diagram	
Type	restriction of xs:float
Facets	minInclusive 0
Used by	Attributes InjectorConfigurationType/@InjectionVolume, OvenConfigurationType/@Temperature, PumpConfigurationType/@FlowRate
Source	<pre><xs:simpleType name="PositiveFloatType"> <xs:restriction base="xs:float"> <xs:minInclusive value="0"/> </xs:restriction> </xs:simpleType></pre>

Simple Type RatioType

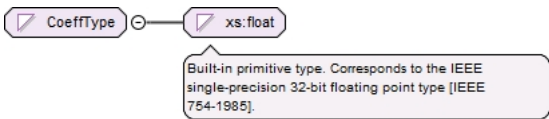
Namespace	No namespace
Annotations	only accepts values between 0-100 iin the form of int:int, user must ensure a valid ratio is input (such as both sides adding to 100)
Diagram	
Type	restriction of xs:string
Facets	pattern (100 [0-9]?[0-9]):(100 [0-9]?[0-9])

Used by	Element EluentType/Ratio
Source	<pre> <xs:simpleType name="RatioType"> <xs:annotation> <xs:documentation>only accepts values between 0-100 iin the form of int:int, user must ensure a valid ratio is input (such as both sides adding to 100)</xs:documentation> </xs:annotation> <xs:restriction base="xs:string"> <xs:pattern value="(100 [0-9]?[0-9]):(100 [0-9]?[0-9])"/> </xs:restriction> </xs:simpleType> </pre>

Simple Type DegreePolyType

Namespace	No namespace				
Annotations	defines if it is a 3rd, 4th or 5th order polynomial				
Diagram					
Type	restriction of xs:int				
Facets	<table> <tr> <td>maxInclusive</td><td>5</td></tr> <tr> <td>minInclusive</td><td>3</td></tr> </table>	maxInclusive	5	minInclusive	3
maxInclusive	5				
minInclusive	3				
Used by	Element CalibrationDataType/TypeOffFit				
Source	<pre> <xs:simpleType name="DegreePolyType"> <xs:annotation> <xs:documentation>defines if it is a 3rd, 4th or 5th order polynomial</xs:documentation> </xs:annotation> <xs:restriction base="xs:int"> <xs:minInclusive value="3"/> <xs:maxInclusive value="5"/> </xs:restriction> </xs:simpleType> </pre>				

Simple Type CoeffType

Namespace	No namespace
Diagram	
Type	xs:float
Used by	Elements CoeffListType/CubicCoeff, CoeffListType/LinearCoeff, CoeffListType/QuadraticCoeff, CoeffListType/QuarticCoeff, CoeffListType/QuinticCoeff, CoeffListType/constant
Source	<pre> <xs:simpleType name="CoeffType"> <xs:restriction base="xs:float"/> </xs:simpleType> </pre>

Attribute(s)

Attribute ContactType / @FirstName

Namespace	No namespace
Type	xs:string
Properties	use: required
Used by	Complex Type ContactType
Source	<pre> <xs:attribute name="FirstName" type="xs:string" use="required"/> </pre>

Attribute ContactType / @MiddleName

Namespace	No namespace
Type	xs:string

Properties	use:	optional
Used by	Complex Type	ContactType
Source	<xs:attribute name="MiddleName" type="xs:string" use="optional"/>	

Attribute ContactType / @LastName

Namespace	No namespace	
Type	xs:string	
Properties	use:	required
Used by	Complex Type	ContactType
Source	<xs:attribute name="LastName" type="xs:string" use="required"/>	

Attribute ContactType / @E-mail

Namespace	No namespace	
Type	xs:string	
Properties	content:	simple
Used by	Complex Type	ContactType
Source	<xs:attribute name="E-mail" type="xs:string"/>	

Attribute ContactType / @Institution

Namespace	No namespace	
Type	xs:string	
Properties	use:	optional
Used by	Complex Type	ContactType
Source	<xs:attribute name="Institution" type="xs:string" use="optional"/>	

Attribute MetadataType / @date

Namespace	No namespace	
Type	xs:date	
Properties	use:	required
Used by	Complex Type	MetadataType
Source	<xs:attribute name="date" type="xs:date" use="required"/>	

Attribute MetadataType / @MethodName

Namespace	No namespace	
Type	xs:string	
Properties	use:	required
Used by	Complex Type	MetadataType
Source	<xs:attribute name="MethodName" type="xs:string" use="required"/>	

Attribute MetadataType / @CalibrationName

Namespace	No namespace	
Type	xs:string	
Properties	use:	optional
Used by	Complex Type	MetadataType
Source	<xs:attribute name="CalibrationName" type="xs:string" use="optional"/>	

Attribute MetadataType / @Operator

Namespace	No namespace
Annotations	If multiple contacts, state name (first-last) of who ran the sample
Type	xs:string
Properties	content: simple
Used by	Complex Type MetadataType
Source	<pre><xs:attribute name="Operator" type="xs:string"> <xs:annotation> <xs:documentation>If multiple contacts, state name (first-last) of who ran the sample</ xs:documentation> </xs:annotation> </xs:attribute></pre>

Attribute MetadataType / @InstrumentName

Namespace	No namespace
Type	xs:string
Properties	use: required
Used by	Complex Type MetadataType
Source	<pre><xs:attribute name="InstrumentName" type="xs:string" use="required"/></pre>

Attribute MetadataType / @RunTime

Namespace	No namespace
Annotations	Length of the run sample in Minutes
Type	xs:time
Properties	use: optional
Used by	Complex Type MetadataType
Source	<pre><xs:attribute name="RunTime" type="xs:time" use="optional"> <xs:annotation> <xs:documentation>Length of the run sample in Minutes</xs:documentation> </xs:annotation> </xs:attribute></pre>

Attribute MetadataType / @lastServiceDate

Namespace	No namespace
Type	xs:date
Properties	use: optional
Used by	Complex Type MetadataType
Source	<pre><xs:attribute name="lastServiceDate" type="xs:date" use="optional"/></pre>

Attribute MetadataType / @SampleName

Namespace	No namespace
Type	xs:string
Properties	use: required
Used by	Complex Type MetadataType
Source	<pre><xs:attribute name="SampleName" type="xs:string" use="required"/></pre>

Attribute InstrumentInfoType / @Name

Namespace	No namespace
Type	xs:string

Properties	use: optional
Used by	Complex Type InstrumentInfoType
Source	<code><xs:attribute name="Name" type="xs:string" use="optional"/></code>

Attribute InstrumentInfoType / @SerialNumber

Namespace	No namespace
Annotations	for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.
Type	xs:string
Properties	content: simple
Used by	Complex Type InstrumentInfoType
Source	<code><xs:attribute name="SerialNumber" type="xs:string"> <xs:annotation> <xs:documentation>for saving what column were used, can be found on providers website, must be an unambiguous string which can be directly linked to the item.</xs:documentation> </xs:annotation> </xs:attribute></code>

Attribute InstrumentInfoType / @Manufacturer

Namespace	No namespace
Annotations	for saving what columns were used
Type	xs:string
Properties	use: required
Used by	Complex Type InstrumentInfoType
Source	<code><xs:attribute name="Manufacturer" type="xs:string" use="required"> <xs:annotation> <xs:documentation>for saving what columns were used</xs:documentation> </xs:annotation> </xs:attribute></code>

Attribute ColumnConfigurationType / @PoreSizeMin

Namespace	No namespace
Type	xs:positiveInteger
Properties	use: required
Used by	Complex Type ColumnConfigurationType
Source	<code><xs:attribute name="PoreSizeMin" type="xs:positiveInteger" use="required"/></code>

Attribute ColumnConfigurationType / @PoreSizeMax

Namespace	No namespace
Type	xs:positiveInteger
Properties	use: required
Used by	Complex Type ColumnConfigurationType
Source	<code><xs:attribute name="PoreSizeMax" type="xs:positiveInteger" use="required"/></code>

Attribute ColumnConfigurationType / @StationaryPhase

Namespace	No namespace
Type	xs:string
Properties	use: optional
Used by	Complex Type ColumnConfigurationType
Source	<code><xs:attribute name="StationaryPhase" type="xs:string" use="optional"/></code>

Attribute ColumnConfigurationType / @DateOfInstallation

Namespace	No namespace
Type	xs:date
Properties	use: optional
Used by	Complex Type ColumnConfigurationType
Source	<code><xs:attribute name="DateOfInstallation" type="xs:date" use="optional"/></code>

Attribute OvenConfigurationType / @Temperature

Namespace	No namespace
Annotations	Must be saved in celcius
Type	PositiveFloatType
Properties	use: required
Facets	minInclusive 0
Used by	Complex Type OvenConfigurationType
Source	<code><xs:attribute name="Temperature" type="PositiveFloatType" use="required"> <xs:annotation> <xs:documentation>Must be saved in celcius</xs:documentation> </xs:annotation> </xs:attribute></code>

Attribute PumpConfigurationType / @FlowRate

Namespace	No namespace
Annotations	must be saved as mL/min
Type	PositiveFloatType
Properties	use: required
Facets	minInclusive 0
Used by	Complex Type PumpConfigurationType
Source	<code><xs:attribute name="FlowRate" type="PositiveFloatType" use="required"> <xs:annotation> <xs:documentation>must be saved as mL/min</xs:documentation> </xs:annotation> </xs:attribute></code>

Attribute InjectorConfigurationType / @InjectionVolume

Namespace	No namespace
Annotations	Must be saved in µL
Type	PositiveFloatType
Properties	use: required
Facets	minInclusive 0
Used by	Complex Type InjectorConfigurationType
Source	<code><xs:attribute name="InjectionVolume" type="PositiveFloatType" use="required"> <xs:annotation> <xs:documentation>Must be saved in µL</xs:documentation> </xs:annotation> </xs:attribute></code>

Attribute RawDataType / @Compression

Namespace	No namespace
Type	xs:boolean
Properties	use: required

Used by	Complex Type RawDataType
Source	<code><xs:attribute name="Compression" type="xs:boolean" use="required"/></code>

Attribute IntergrationValuesType / @StartVolume

Namespace	No namespace
Annotations	Must be saved in mL
Type	xs:float
Properties	use: required
Used by	Complex Type IntergrationValuesType
Source	<code><xs:attribute name="StartVolume" type="xs:float" use="required"> <xs:annotation> <xs:documentation>Must be saved in mL</xs:documentation> </xs:annotation> </xs:attribute></code>

Attribute IntergrationValuesType / @EndVolume

Namespace	No namespace
Annotations	Must be saved in mL
Type	xs:float
Properties	use: required
Used by	Complex Type IntergrationValuesType
Source	<code><xs:attribute name="EndVolume" type="xs:float" use="required"> <xs:annotation> <xs:documentation>Must be saved in mL</xs:documentation> </xs:annotation> </xs:attribute></code>

Attribute gpcMLType / @Version

Namespace	No namespace
Annotations	gpcML version used.
Type	xs:string
Properties	use: required
Used by	Complex Type gpcMLType
Source	<code><xs:attribute name="Version" type="xs:string" use="required"> <xs:annotation> <xs:documentation>gpcML version used.</xs:documentation> </xs:annotation> </xs:attribute></code>

Attribute gpcMLType / @id

Namespace	No namespace
Annotations	id associated with the gpcML document
Type	xs:int
Properties	content: simple
Used by	Complex Type gpcMLType
Source	<code><xs:attribute name="id" type="xs:int"> <xs:annotation> <xs:documentation>id associated with the gpcML document</xs:documentation> </xs:annotation> </xs:attribute></code>

Attribute Sample / @Name

Namespace	No namespace
Type	xs:string

Properties	use:	required
Used by	Complex Type	Sample
Source	<xs:attribute name="Name" type="xs:string" use="required"/>	

Attribute ColumnInfoType / @DateOfInstallation

Namespace	No namespace	
Type	xs:date	
Properties	use:	optional
Used by	Complex Type	ColumnInfoType
Source	<xs:attribute name="DateOfInstallation" type="xs:date" use="optional"/>	

Attribute ColumnInfoType / @PlateCount

Namespace	No namespace	
Type	xs:int	
Properties	use:	optional
Used by	Complex Type	ColumnInfoType
Source	<xs:attribute name="PlateCount" type="xs:int" use="optional"/>	

Attribute GuardColumnConfigurationType / @ParticleSize

Namespace	No namespace	
Type	xs:float	
Properties	use:	optional
Used by	Complex Type	GuardColumnConfigurationType
Source	<xs:attribute name="ParticleSize" type="xs:float" use="optional"/>	

Attribute GuardColumnConfigurationType / @DateOfInstallation

Namespace	No namespace	
Type	xs:date	
Properties	content:	simple
Used by	Complex Type	GuardColumnConfigurationType
Source	<xs:attribute name="DateOfInstallation" type="xs:date"/>	